

A Technical Introduction to Cross entropy loss

$$H(p, q) = -\sum_x p(x) \log q(x)$$

Section 1

$$\hat{y}_i = g(\mathbf{w} \cdot \mathbf{x}_i) = \frac{1}{1 + \exp(-\mathbf{w}_0 - \mathbf{w}_1 x_{i1} - \dots - \mathbf{w}_p x_{ip})}, \quad \{\hat{y}_i\} = g(\mathbf{w} \cdot \mathbf{x}_i)$$

Section 2

$$\frac{\partial L}{\partial \mathbf{w}_0} = -\sum_{i=1}^N [y_i(1 - y_i) - (1 - y_i)y_i] = -\sum_{i=1}^N [y_i - y_i^2] = \sum_{i=1}^N x_{i0}(y_i - y_i^2)$$

Section 3

In information theory, the cross-entropy between two probability distributions p and q , over the same underlying set of events, measures the average number of bits needed to identify an event drawn from the set when the coding scheme used for the set is optimized for an estimated probability distribution q , rather than the true distribution p .

Section 4

$$J(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N H(p_i, q_i) = -\frac{1}{N} \sum_{i=1}^N [y_i \log y_i + (1 - y_i) \log (1 - y_i)]$$

Section 5

$$\frac{\partial}{\partial \mathbf{w}_j} \ln \hat{y}_i = \frac{\partial}{\partial \mathbf{w}_j} \ln \frac{1}{1 + e^{-\mathbf{w}_j x_{ij} + k_j}} = x_{ij} e^{-\mathbf{w}_j x_{ij} + k_j} / (1 + e^{-\mathbf{w}_j x_{ij} + k_j})^2 = x_{ij} (1 - \hat{y}_i)$$

Section 6

where e^k is the cost function of the k th classifier, q^k is the output probability of the k th classifier, p is the true probability to be

estimated, and λ is a parameter between 0 and 1 that defines the 'diversity' that we would like to establish among the ensemble. When $\lambda = 0$ we want each classifier to do its best regardless of the ensemble and when $\lambda = 1$ we would like the classifier to be as diverse as possible.

Section 7

where $H(p)$ is the entropy of p . For discrete probability distributions p and q with the same support X , this means

Section 8

Put together, we obtain the desired result. Note that here we used the natural logarithm $\ln$ instead of $\log$ (canonically $\log_2$) in the loss function L . This merely changes the result by a factor of $\log(e)$ though. Moreover, for all i we reused $k_0 := -\sum_{l=1}^p \mathbf{w}_l x_{il}$ and $k_j := -\mathbf{w}_0 - \sum_{l \neq j} \mathbf{w}_l x_{il}$ as constants w.r.t. each $\mathbf{w}_j$.

Section 9

$$-\int X P(x) \log Q(x) dx = E_p [-\log Q],$$

Section 10

Having set up our notation, $p \in \{y, 1 - y\}$ and $q \in \{\hat{y}, 1 - \hat{y}\}$, we can use cross-entropy to get a measure of dissimilarity between p and q :

Section 11

$$e^k = H(p, q^k) - \lambda \sum_{j \neq k} H(q^j, q^k)$$

Section 12

Relation to linear regression The gradient of the cross-entropy loss for logistic regression is equal to the gradient of the squared-error loss for linear regression (up to a constant factor). To see this, define

Section 13

where N is the size of the test set, and $q(x)$ is the probability of event x estimated from the training set. In other words, $q(x_i)$ is the probability

estimate of the model that the i -th word of the text is x_i $\{\displaystyle x_i\}$. The sum is averaged over the N $\{\displaystyle N\}$ words of the test. This is a Monte Carlo estimate of the true cross-entropy, where the test set is treated as samples from $p(x)$ $\{\displaystyle p(x)\}$.

Section 14

$$\partial \partial w \ln \blacksquare (1 - y^{\wedge} i) = \partial \partial w \ln \blacksquare [1 - 1 + e - w j x i j + k j] = -x i j 1 + e - w j x i j + k j = -x i j y^{\wedge} i \{ \displaystyle \frac{\{\partial \partial w\}}{\{\partial \partial w\}} \ln (1 - \{\hat{y}\}_i) = \frac{\{\partial \partial w\}}{\{\partial \partial w\}} \ln \left(1 - \frac{1}{1 + e^{\{-w\}_j x_{\{i\}} + k_{\{j\}} \}} \right) = \frac{\{-x_{\{i\}}\} \{1 + e^{\{-w\}_j x_{\{i\}} + k_{\{j\}} \}}}{\{-x_{\{i\}}\} \{1 + e^{\{-w\}_j x_{\{i\}} + k_{\{j\}} \}}} = -x_{\{i\}} \{\hat{y}\}_i \}$$

Section 15

Since the logarithm is a monotonically increasing function, the maximizing value of θ is unaffected by this final step. Similarly, the maximizing value of θ is unaffected by the factor of N . So we observe that the likelihood maximization amounts to minimization of the cross-entropy.

Section 16

Cross-entropy loss function and logistic regression

Cross-entropy can be used to define a loss function in machine learning and optimization. Mao, Mohri, and Zhong (2023) give an extensive analysis of the properties of the family of cross-entropy loss functions in machine learning, including theoretical learning guarantees and extensions to adversarial learning. The true probability p_i is the true label, and the given distribution q_i is the predicted value of the current model. This is also known as the log loss (or logarithmic loss or logistic loss); the terms "log loss" and "cross-entropy loss" are used interchangeably. More specifically, consider a binary regression model which can be used to classify observations into two possible classes (often simply labelled 0 and 1). The output of the model for a given observation, given a vector of input features x , can be interpreted as a probability, which serves as the basis for classifying the observation. In logistic regression, the probability is modeled using the logistic function $g(z) = 1 / (1 + e^{-z})$ where z is some function of the input vector x , commonly just a linear function. The probability of the output $y = 1$ is given by